

```
In [1]: 1 #Add or remove libraries as needed
2 import cv2
3 import numpy as np
4 import matplotlib.pyplot as plt
5 import matplotlib.cm as cm
6
```

Task#1: Load/read cat.jpeg as a grayscale image [1 mark]

```
In [42]: 1 img1 = cv2.imread("cat.jpeg", cv2.IMREAD_GRAYSCALE)
```

Task#2: Apply thresholding (thresh=100) to the image and display the results [1.5 mark]

Note: use cmap = 'gray' to properly display the thresholded image

```
In [44]: 1
2 ret, thresholded_image = cv2.threshold(img1, 100, 255, cv2.THRESH_BINARY)
3
4 plt.figure(figsize=(12, 12))
5 plt.subplot(1, 2, 1)
6 plt.imshow(img1, cmap='gray')
7 plt.title('Original')
8 plt.axis('off')
9
10
11 plt.subplot(1, 2, 2)
12 plt.imshow(thresholded_image, cmap='gray')
13 plt.title('Thresholded Image (thresh=100)')
14 plt.axis('off')
15 plt.show()
```

Original



Thresholded Image (thresh=100)



Task#3: Load/read the camera man, blur it using a 9x9 gaussian filter (sigma=5), and display the results. [2.5 marks]

```
In [45]: 1 img2 = cv2.imread("cameraman.tif")
2 img2_Blure = cv2.GaussianBlur(img2, (5, 5), 5)
3
4 plt.figure(figsize=(12, 12))
5 plt.subplot(1, 2, 1)
6 plt.imshow(img2)
7 plt.title('Original')
8 plt.axis('off')
9
10
11 plt.subplot(1, 2, 2)
12 plt.imshow(img2_Blure)
13 plt.title('Image wiht 9x9 gaussian filter (sigma=5)')
14 plt.axis('off')
15 plt.show()
```

Original



Image wiht 9x9 gaussian filter (sigma=5)



```
In [ ]: 1
```